UNIT ONE
LOGIC
GAC 135

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LOGIC

0.1 Introduction

To understand mathematics and mathematical arguments, it is necessary to have a solid understanding of logic and the way in which known facts can be combined to prove facts.

- The rules of logic give precise meaning to mathematical statements.
- The rules are used to distinguish between valid and invalid mathematical arguments.
- In Mathematics ,logic refers to sentential calculus (also known as **propositional calculus**, which is a logical and well-defined system based on sentences.)
- A **sentence** is a well-formed expression within a language and a syntax (the arrangement of words and phrases to create well - formed sentences in a language) that can be evaluated as true or false using algebraic tools. For example: The sentence "The ball is blue " can be evaluated as either true or false, but the sentence " Red is the best colour ", cannot be.

- The language of mathematics consists primarily of declarative sentences. If a sentence can be classified as true or false ,it is called a **statement**. The truth or falsity of a statement is known as its **truth value.**
- A proposition is any meaningful statement that is true or false ,but not both.
- Lowercase letters,such as $p, q, r...$, would be used to represent propositions,but for compound propositions,uppercase letters,such as $P, Q, R...$,would be used.
- The notation;

$$p : 1 + 1 = 3$$

  would be used to define p as the proposition 1+1=3
- The Truth Value (Logical Value) of a proposition is true ,denoted by T, if it is a true statement and false , denoted by F,if it is a false statement.

1. A **sentence** where truth depends on the value of one or more variables is called **open sentence**

Example 1.1

Consider the following sentences, which of them is a statement, not a statement or an open sentence

1. $2 + 2 = 4$
2. Help me, please.
3. Read this carefully.
4. $x - y = y - x$
5. The number 2 is positive and the number 3 is negative.
6. $x^2 - 5x + 6 = 0$
7. $x + y = y + x$ for all $x, y \in \mathbb{R}$
8. $2 + 3 = 7$
9. x is greater than 5

Example 1.2

Which of the sentences are statements, not a statement or an open sentence

1. If x is a real number, then $x^2 \geq 0$
2. Seven is a prime number.
3. Seven is an even number.
4. This sentence is false.
5. $2^n = n^2$ for some $n \in \mathbb{N}$.
6. It is not true that 3 is an even integer or 7 is a prime.
7. If the world is flat ,then $2 + 2 = 4$
8. $A^2 = 0$ inputs $A = 0$ for all A.
9. The set of integers $\mathbb{Z}$.
10. What is the solution of $2x = 84$?
11. $cosx = -1$

1.2 PROPOSITIONAL LOGIC AND RELATED CONCEPTS

Propositions can be combined to form new sentences (compound sentences) using the connectives: NOT, OR , AND, IF-THEN,and IF AND ONLY IF.

Example 1.3

The propositions:

1. The number 2 is even .

2. The number 3 is odd.
   can be combined to form:
   1. The number 2 is not even or (The number 2 is odd) .
   2. The number 2 is even or the number 3 is odd.

1.3 Logical connectives and Truth Tables

The language of logic includes a set of variables and a set of **connectives** from which any sentences can be built according to the rules of the syntax.
New propositions,called compound propositions or propositional functions, can be obtained from old ones(atomic proposition)p, q, r, s, t,by using symbolic connectives (sentential connectives or logical operations).

Negation ("Not" connective)

Let p stand for a given statement.
Then $\sim p$ (read not p) represents the logical opposite (negation) of p.
When p is true, $\sim p$ is false; When p is false, $\sim p$ is true.
This can be summarized in a truth table:

p	$\sim p$
T	F
F	T

Where T stands for true and F stands for false.

Note

Negation is one of the fundamental logical connectives used to construct compound propositions. It is unary operator.

Example 1.4

Let p, q and r be given as follows:

p: $y \geq x$,

q: Five is an even number,

r: The set of integers is countable.

Then their negations can be written as:

- $\sim$ p: y $< x$
- $\sim$ q :Five is not an even number or , Five is an odd number.
- $\sim$ r:The set of integers is not countable or, the set of integers is uncountable.

Disjunction("OR connective)

Let p and q be propositions.
The disjunction of p and q,denoted $p \vee q$,is the proposition : p or q.
The "or" is used in an inclusive way.
This proposition is false only when both p and q are false; otherwise, it is true.

Example 1.5

Let p and q be given as follows :

p: The set S is compact.
q: The set S is convex.

Then the disjunction of p and q denoted can be written as ;

$p \vee q$:The set S is compact or convex.

Conjunction("AND" connective)

The **conjuction** pf p and q, denoted $p \wedge q$, is the proposition :p and q. This proposition is defined to be true only when both p and q are true ; otherwise, it is false.

Example 1.6

Using some statement in Eg. 1.5, the conjunction of p and q can be written as The set S is compact and convex

Conditional (Implication)("IF,THEN",connective)

Let p and q be propositions.
The implication $p \Longrightarrow q$ is the proposition that is false only when p is true and q is false; otherwise, it is true .
p is called the **hypothesis**(an antecedent) and q is called the **conclusion**(consequent).

It is important to recognize that in mathematical writing , the conditional statement can be disquised in several equivalent forms.
The following expressions all mean exactly the same thing:

- If p,then q
- p implies q
- p only if q
- q if p
- Whenever p, then also q.
- For q, it is sufficient that p
- For p, it is necessary that q
- q provided that p
- q whenever p.
- p is sufficient condition for q.
- q is necessary condition for p .

- If p, q
- A necessary condition for p is q
- q unless $\sim p$
- a sufficient condition for p is q
- q follows from p

Example 1.7

Identify the antecedent and the consequent in each of the following statements.

1. If n is an integer, then $2n$ is an even number .
2. A square matrix A is invertible provided that its determinant is not zero
3. A series converges whenever it converges absolutely.
4. Continuity is a necessary condition for differentiablility.
5. A function is integrable provided the function is continuous.
6. Whenever people agree with me I feel I must be wrong .

- The **Converse** of $p \to q$ is the proposition $q \to p$.
- The **Opposite** or **Inverse** of $p \to q$ is the proposition $\sim p \to \sim q$.
- The **Contrapositive** of $p \to q$ is the proposition $\sim q \to \sim p$.

But sometimes,if p and q are just the right statements, it can happen that $p \longrightarrow q$ and $q \longrightarrow p$ are both necessarily true.

Biconditional("IF AND ONLY IF" connective)

$p \longrightarrow q$ is not the same as the $q \longrightarrow p$.
The **converse** of $p \longrightarrow q$ is $q \longrightarrow p$, so a conditional statement and its converse express entirely different things.

Example 1.8

Let p and q be given as follows:

p: x is a multiple of 6

q: x is divisible by 2.

Find $p \to q$ and the converse and truth value .

Example 1.9

Let p and q be given as follows:

p: x is even

q: x is divisible by 2.

Find $p \to q$ and the converse and truth value .

- The **biconditional** proposition of p and q, denoted by $p \leftrightarrow q$, is the propositional function that is true when both p and q have the same truth values and false if p and q have opposite truth values.
  The connective $\leftrightarrow$ is called the **equivalence** connective.
  Also reads,"p if and only if q or " p is a necessary and sufficient condition for q.
- p if and only q.
- p is a necessary and sufficient condition for q.
- For p it is necessary and sufficient that q.
- If p,then q, and conversely.

Example 1.10

Without changing their meanings, convert each of the following sentences into a sentence having the form "p if only if q".

1. For matrix A to be invertible, it is necessary and sufficient that $det(\mathrm{A}) \neq 0$.
2. If a function has a constant derivative then it is linear ,and conversely.
3. $xy = 0$ then $x = 0$ or $y = 0$, and conversely.
4. If $a \in \mathbb{Q}$ then $5a \in \mathbb{Q}$, and if $5a \in \mathbb{Q}$ then $a \in \mathbb{Q}$.

Tautology

Definition:
A compound proposition is called a **tautology** if it is always true , regardless of the truth values of the basic propositions which comprise it.

Definition:
A proposition, which is always false, no matter what the truth values are assigned to its component , is called **contradicton**. A contradicton is analogous to tautology.

The compound propositions P and Q are called logically equivalent if $P \iff Q$ is a tautology. The notation $P \equiv Q$ denotes that P and Q are logically equivalent.

Example 1.11

Use the truth table to show that the following are logically equivalent

1. $\sim (p \vee q)$ and $\sim p \wedge \sim q$
2. $p \rightarrow q$ and $\sim p \vee q$

3. $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$
4. $p \vee (p \wedge q)$ and p
5. $p \wedge (p \vee q)$ and p

Example 1.12

Prove that the following are logically equivalent.

1. $\sim (p \rightarrow q)$ and $p \wedge \sim q$
2. $\sim (p \vee (\sim p \wedge q))$ and $\sim p \wedge \sim q$
3. $(p \wedge q) \rightarrow (p \vee q)$

Exercises

1. Mark each statement True or False. Justify each answer.
   a. In order to be classified as a statement, a sentence must be true.
   b. Some statements are both true and false.
   c. When statement p is true , its negation $\sim p$ is false.
   d. A statement and its negation may both be false.
   e. In mathematical logic, the word "or "has an inclusive meaning.
2. Mark each statement True or False. Justify each answer.
   a. In an implication $p \Rightarrow q$,statement p is referred to as the proposition.
   b. The only case where $p \Rightarrow q$ is false when p is true and q is false.
   c. " If p, then q" is equivalent to "p whenever q".
   d. The negation of a conjunction is the disjunction of the negations of the individual parts.
   e. The negation of $p \Rightarrow q$ is $q \Rightarrow p$.

3. Write the negation of each statement.
    a. The 3×3 identity matrix is singular.
    b. The function $f(x) = \sin x$ is bounded on $\mathbb{R}$
    c. The functions f and g are linear.
    d. Six is prime or seven is odd.
    e. if x is in D, then $f(x) < 5$
    f. If (a_n) is monotone and bounded ,then (a_n) is convergent.
    g. If f is injective, then S is finite or denumerable.

4. Write the negation of each statement.
    a. The function $f(x)$= x^2 - 9 is continuous at $x = 3$.
    b. The relation R is reflexive or symmetric .
    c. Four and nine are relatively prime.
    d. x is in A or X is not in B.
    e. If $x < 7$, then f(x) is not in C.
    f. If (a_n) is convergent, then (a_n) is monotone and bounded.
    g. If f is continuous and A is open, then $f^{-1}(A)$ is open .

5. Identify the antecedent and the consequent in each statement.
    a. M has a zero eigenvalue whenever M is singular.
    b. Linearity is a sufficient condition for continuity.
    c. A sequence is Cauchy only ‘if it is bounded.
    d. $x <$ provided that $y > 5$.
6. Identify the antecedent and the consequent in each statement.
    a. A sequence is convergent if it is Cauchy.
    b. Convergence is a necessary condition for boundedness.
    c. Orthogonality implies invertability.
    d. K is closed and bounded only if K is compact.
7. Construct a truth table for each statement.
    a. $p \Rightarrow \sim q$
    b. $[p \wedge (p \Rightarrow q)] \Rightarrow q$
    c. $[p \Rightarrow (q \wedge \sim q)] \Leftrightarrow \sim p$

8. Construct a truth table for each statement.
    a. $p \vee \sim q$
    b. $q \wedge \sim p$
    c. $[(\sim q) \wedge (p \Rightarrow q)] \Leftrightarrow \sim p$
9. Indicate whether each statement is true or false.
    a. $3 \leq 5$ and 11 is odd.
    b. $3^2 = 8$ or $2 + 3 = 5$.
    c. $5 > 8$ or 3 is even.
    d. If 6 is even, then 9 is odd.
    e. If $8 < 3$, then $2^2 = 5$.
    f. If 7 is odd, then 1 is prime.
    g. If 8 is even and 5 is not prime, then $4 < 7$.
    h. If 3 is odd or $4 > 6$, then $9 \leq 5$.
    i. If both $5 - 3 = 2$ and $5 + 3 = 2$, then $9 = 4$.
    j. It is not the case that 5 is even and 7 is prime.

10. Indicate whether each statement is true or false.
    a. $2+3=5$ and 5 is even.
    b. $3+4=5$ or $4+5=6$.
    c. 7 is even or 6 is not prime.
    d. If $4+4=8$, then 9 is prime.
    e. If 6 is prime, then $8<6$.
    f. If $6<2$, then $4+4=7$.
    g. If 8 is prime or 7 is odd, then 9 is even.
    h. If $2+5=7$ only if $3+4=8$, then $3^2=9$.
    i. If both $5-3=2$ and $5+3=8$, then $8-3=4$.
    j. It is not the case that 5 is not prime and 3 is odd.
11. Determine whether the following are tautology.
    a. $(\sim p \wedge (p \rightarrow q)) \rightarrow \sim q$
    b. $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$
12. Show that the following are logically equivalent.
    a. $p \leftrightarrow q$ and $(p \wedge q) \vee (\sim p \wedge \sim q)$
    b. $\sim (p \leftrightarrow q)$ and $p \leftrightarrow \sim q$

PROPOSITIONS AND QUANTIFIERS

Statements such as $x > 3$ are often found in mathematical assertions. These statements are not propositions when the variables are not specified. However, one can produce propositions from such statements.

- **Predicate**: part of a sentence that attribute a property to the subject.

A **predicate** is an expression involving one ore more variables defined on some domain, called the **domain of discourse**.

OR

A **predicate** calculus is the augmented system of symbols and rules.

OR

Let $U_1, U_2, \ldots, U_n$ be nonempty sets. An **n-place predicate** over $U_1 \times U_2 \times, \ldots, \times U_n$ is a function $p(x_1, x_2, \ldots, x_n)$ with domain $U_1 \times U_2 \times, \ldots, \times U_n$ which has propositions as its function values.

The statement;

"x is greater than 3"

has two parts.
The first part:

the variable "x"

is subject to the statement.

The second part - The **predicate**,

"is greater than 3"

refers to a property that the subject of the statement can have.

We can denote the statement

"x is greater than 3"

by $p(x)$,where p denotes the predicate

"x is greater than 3"

and x is the variable.

The statement $p(x)$ is also said to be the value of the **propositional function** at x.
Once a value has been assigned to the variable x, the statement $p(x)$ becomes a proposition and has a truth value

Substitution of a particular value for the variable(s) produces a proposition, which is either true or false.
For instance,$p(n)$: n is prime is a predicate on the natural numbers.
Observe that $p(1)$ is false ,$p(2)$ is true .
In the expression $p(x)$, x is called **free variable**.

As x varies, the truth value of $p(x)$ varies as well.
The set of true values of a predicate $p(x)$ is called the truth set and will be denoted by T_p .

An expression involving two variable (2 - place predicate over $\mathbb{N} \times \mathbb{N}$)
$p : (m, n) ; n > 2^m$.

We found that the sentence

$$x^2 - 5x + 6 = 0$$

needed to be considered within a particular context in order to become a statement.
When a sentence involves a variable such as x , it is customary to use functional notation when referring to it.
Thus, we write

$$p(x)\colon\ x^2 - 5x + 6 = 0$$

to indicate that $p(x)$ is sentence $x^2 - 5x + 6 = 0$.
For a specific value of x,$p(x)$ becomes a statement that is either true or false.
For example, $p(2)$ is true and $p(4)$ is false.

Example 1.14

1. Let $p(x)$ denote the statement "$x > 3$".
   What are the truth values of $p(4)$ and $p(2)$?
2. Let $p(x, y)$ denote the statement "$x = y + 3$".
   What are the truth values of the propositions $p(1, 2)$ and $p(3, 0)$?
3. Let $p(x, y, z)$ denote the statement "$x + y = z$".
   What are the truth values of the propositions $p(1, 2, 3)$ and $p(0, 0, 1)$?

- Given a predicate p, and a domain D,how often does the predicate become true?
- Some of the possible answers are "always","sometimes" and "never".
- More generally, we want to know what **quantity** of cases the predicate becomes true.
- Certain words, symbols or groups of words that help to answer such a question are called **quantifiers**.
- **Quantifiers** tell us something about **"quantifier"**.
- They help us to decide the frequencies with which a predicate becomes true, whether it is satisfied by:

  - **no elements**
  - **one element**
  - **some elements**
  - **all elements**

  of a domain.

- The common quantifiers are :
  The **existential quantifier** ($\exists$) is read, "These exists....",This is at least one...", or something equivalent.
  If $p(x)$ is the proposition, then symbolically:
  $$\exists \mathrm{x} \in \mathrm{D} \ni p(x) \quad \text{or} \quad \exists \mathrm{x} \ni p(x)$$
  The notation
  $$\exists\ \mathrm{x} \in \mathrm{D} \ni p(x)$$
  is a proposition that is true if there is at least one value of $\mathrm{x} \in \mathrm{D}$ where $p(x)$ is true; otherwise it is false.
  The symbol is just a shorthand notation for the phrase "such that"
- The **universal quantifier** ($\forall$) is read , " For every ..." , "For all...","For each...", or a similar phrase.
  If $p(x)$ is the proposition, then symbolically:
  $$\forall\ \mathrm{x} \in \mathrm{D},\ p(x) \quad \text{or} \quad \forall\ \mathrm{x},\ p(x)$$
  The notation
  $$\forall\ \mathrm{x} \in \mathrm{D},\ p(x)$$
  is a proposition that is true if all values of $\mathrm{x} \in \mathrm{D}$ where $p(x)$ is true; otherwise it is false.

Example 1.15

- The sentence
For every x, x^2 - $5x + 6 = 0$ is a statement since it is false symbolically: $\forall x \in \mathbb{R}$, x^2 - $5x + 6 = 0$.

$$\text{Or } \forall x \in \mathbb{R}, p(x) \text{ or } \forall x, p(x)$$

The sentence there exists an x such that x^2 - $5x + 6 = 0$ is also a statement and it is true.

- When a quantifier is placed in front of a variable, we say that the variable is **quantified**.
- If $p(x)$ is a proposition which is true then $p(x)$ is true for all values of x in the domain of $p(x)$.
For example, if k is a non negative integer, then the predicate $p(k)$: $2k$ is even is true for all $k \in \mathbb{N}$.we write $\forall k \in \mathbb{N}$,$2k$ is even.

For example:
The statement
There exist a number less than 7 can be written

$$\exists\, x \ni x < 7$$

or in the abbreviated form

$$\exists\, x < 7$$

where it is understood that x represents a real number.
Sometimes the quantifier is not explicitly written down, as in the statement:

If x is greater than 1,then x^2 is greater than 1.

The intended meaning is:

$$\forall x \text{ if } x > 1 \text{ then } x^2 > 1.$$

In general, if a variable is used in the antecedent of an implication without being explicitly quantified, the **universal quantifier is assumed to apply**.

The words "no" and "none" are universal quantifiers in a negative sentence. For example ,"None of them are $p(x)$" is equivalent to "All of them, are not $p(x)$".

Example 1.16

Rewrite each statement using ∃ ,∀, and ∃,as appropriate :

1. There exists a positive number x such that $x^2 = 5$
2. For every positive number M,there is a positive number N such that $N < \frac{1}{M}$.
3. If $n \geq N$,then $|f_n(x) - f(x)| \leq 3$ forall x in A.
4. No positive number x satisfies the equation $f(x) = 5$.

- The proposition

$$\forall x \in D,\ p(x)$$

is false if $p(x)$ is false for at least one value of x.
In this case x is called a **counterexample**.

For example, let $p(x)$ mean $x^2 \geq x$.
Let D_1 and D_2 be two domains of interpretation where $D_1 \in \mathbb{N}$ and $D_2 \in \mathbb{R}$.
Then the statement $\forall x,\ p(x)$ is true in D_1,but false in D_2 since $\frac{1}{2} \in D_2$ and $(\frac{1}{2})^2 = \frac{1}{4} < \frac{1}{2}$.

For example, let $p(x)$ mean $x^2 = -x$.
Let $D_1 = \mathbb{N}$ and $D_2 = \mathbb{Z}$.Then the statement

$$\exists x \ni p(x)$$

is false in D_1, but true in D_2, since -1 $\in D_2$ and $(-1)^2 = 1 = -(-1)$.

Example 1.17

1. Show that the proposition $\forall x \in \mathbb{R},\ x > \frac{1}{x}$ is false
2. Write in the form $\forall\ x \in D,\ p(x)$ the proposition, "every real number is either positive , negative, or 0".

- The proposition $\forall x \in D,\ p(x) \to q(x)$ is called the **universal conditional proposition**.

For example, the proposition

$\forall x \in \mathbb{R}$, if $x > 2$ then $x^2 > 4$ is a universal conditional proposition.

Example 1.18

Rewrite the proposition "if a real number is an integer then it is a rational number" as a universal conditional proposition.

- Also often used in predicate logic is the unique existential quantifier $\exists!$ which means " there exists a unique..."

$$\exists!\ x \ni p(x) \text{ or } \exists!\ x \in D \ni p(x)$$

- Other phrases for uniqueness qualification include :

- "there is exactly one"
- "there is one and only one".

Example 1.19

Which of the following statements are true and which are false:

a. $\exists!\ x \in \mathbb{R} \ni \forall y \in \mathbb{R}, xy = y$.
b. $\exists!$ integer x such that $\frac{1}{x}$ is an integer.

Negative of Quantified Statements

Let $p(x)$ be a proposition and the symbol "$\sim$" represents negation, then

$$\sim [\forall \text{ x}, \; p(x)] \Leftrightarrow [\exists \text{ x} \ni \sim p(x)]$$

Similarly,

$$\sim [\exists \text{ x} \ni p(x)] \Leftrightarrow [\forall \text{ x}, \sim p(x)]$$

For example

1. Statement: The square of every real number is non-negative.

   $$\forall x \in \mathbb{R}, \; x^2 \geq 0$$
   $$\text{Negation: } \exists x \in \mathbb{R} \ni x^2 < 0$$

   There exist a real number whose square is negative.

2. Statement: For every x in A, $f(x) > 5$

   $\forall x$ in A, $f(x) > 5$

   Negation: $\exists x$ in $A \ni f(x) \leq 5$.

   There is an x in A such that $f(x) \leq 5$

- C. Statement: There exists a positive number x such that $< f(x) \leq 1$

  $\exists x > 0 \ni 0 < f(x) \leq 1$

  Negation: $\forall\ y > 0,\ f(x) \leq 0$ or $f(x) > 1$.

  For every positive number x, $f(x) \leq 0$ or $f(x) \geq 1$.

- D. Statement: If a $\in \mathbb{Z}$ is odd then a^2 is odd

  $\forall\ a \in \mathbb{Z},\ a = 2k + 1\ \forall\ k \in \mathbb{Z} \Rightarrow a^2 = 2m + 1\ \forall m \in \mathbb{Z}$. Negation $\exists\ a \in \mathbb{Z} \ni a = 2k + 1\ \forall k \in \mathbb{Z}$ and $a^2 \neq 2m + 1\ \forall m \in \mathbb{Z}$

Example 1.20

Rewrite each statement as Negative quantifier using $\exists$,$\forall$, and $\exists$,as appropriate :

1. There exists a positive number x such that $x^2 = 5$
2. For every positive number M,there is a positive number N such that $N < \frac{1}{M}$.
3. If $n \geq N$,then $|f_n(x) - f(x)| \leq 3$ forall x in A.
4. No positive number x satisfies the equation $f(x) = 5$.

Example

- Assume that the domain for the variables x and y consists of all real numbers. The statement:

$$\forall x \, \forall y \, , x + y = y + x$$

says that $x + y = y + x$ for all real numbers x and y. This is the commutative law for the addition of real numbers.

- Likewise, the statement

$$\forall x \, \forall y \; x + (y + z) = (x + y) + z$$

is the Associative law for the addition of real numbers.
Similarly, the statement

$$\forall x \, \exists y \;\ni\; x + y = 0$$

is the Additive inverse property for the addition of real numbers.

Example 1.21

Translate into English the statement $\forall x \, \forall y, \, x > 0 \wedge y < 0 \rightarrow xy < 0$.

The order of Quantifiers

The order of quantifiers is important,unless all the quantifiers are universal quantifiers or all are existential quantifiers.

Nested

Let $p(x, y)$ be a predicate involving two variables x and y. Then both variables can be quantified. This gives eight possibilities:

(a) (i) $\exists x \; \exists y \ni p(x, y)$
(ii) $\exists y \; \exists x \ni p(x, y)$

(b) (i) $\forall x \forall y, \quad p(x, y)$
(ii) $\forall y, \forall x \quad p(x, y)$

(c) (i) $\forall x \; \exists y \ni p(x, y)$
(ii) $\exists y \ni \forall x, \; p(x, y)$

(d) (i) $\exists x \ni \forall y, \; p(x, y)$
(ii) $\forall y, \; \exists x \ni p(x, y)$

- If the same quantifier is used twice, then the order does not matter, so a)(i) and (ii) have the same meaning as each other, and b)(i) and (ii) have the same meaning as each other.

- But if there is a mixture of $\exists$ and $\forall$, then the order does matter. A look at c)(i) means that for each x that we specify, there exists a y which may depend on x such that $P(x, y)$ is true.
  So changing x may change y. By contrast, according to (c)(ii), there is some constant y such that no matter what value for x, $P(x, y)$ is true.

Similarly, parts (i) and (ii) in (d) are not the same.

For Example (On same quantifiers)

Let $p(x, y)$ be the statement $x + y = y + x$,where $\mathbb{R}$ is the domain

The statement:

$$\forall x\ \forall y, p(x, y)$$

has the same meaning as the statement

$$\forall y\ \forall x, p(x, y)$$

and both are true.

For Example(Mixture of a quantifiers)

Let $p(x, y) : x < y$. Let the domain D be the set of all real numbers. Then $\forall x$, $\exists y \ni p(x, y)$ is true. To see this we rewrite $\exists y \ni \forall x$.

For each x we find a value of y such that $x < y$.
This means finding a number y which is bigger than x.
We can choose $y = x + 1$. However, the statement $\exists y \ni \forall x$, $P(x, y)$, which makes the claim that there is some constant y which is greater than every real number x.
But there is no greatest real number, so the statement is false.

For example, the statement (the domain of discourse is $\mathbb{R}$)

$$\forall x \, \exists y \ni y > x$$

is true.

That is, given any real number x, there is always a real number y that is greater than that x. But the statement

$$\exists y \ni \forall x, \; y > x$$

is false, since there is no fixed real number y that is greater than every real number.

Quantification of Two Variables

Statement	When True?	When False?
$\exists x \exists y \ni p(x, y)$ $\exists y \exists x \ni p(x, y)$	There is a pair x, y for which $p(x, y)$ is true.	$p(x, y)$ is false for every pair x, y
$\forall x \forall y, p(x, y)$ $\forall y \forall x p(x, y)$	$p(x, y)$ is true for every pair x, y	There is a pair x, y for which $p(x, y)$ is false
$\forall x, \exists y \ni p(x, y)$	For every x there is a y for which $p(x, y)$ is true	There is an x such that $p(x, y)$ is false for every y.
$\exists x \ni \forall y, p(x, y)$	There is an x for which $p(x, y)$ is true for every y.	For every x there is a y for which $p(x, y)$ is false.

Example 1.22

Determine the truth value of each statement. Assume that x and y are real numbers. Justify your answer.

1. $\forall x\ \exists y \ni x + y = 0$
2. $\exists y \ni \forall x,\ x + y = 0$
3. $\exists x \ni \forall y,\ x + y \neq 0$
4. $\forall x\ \forall y,\ \exists z \ni x + y = z$
5. $\exists z \ni \forall x\ \forall y,\ x + y = z$

Negating Nested Quantifiers

- Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier.

Example 1.23

Express the negation of the following statements:

1. $\forall x,\ \exists y \ni xy = 1$
2. $\forall \varepsilon > 0,\ \exists \delta > 0 \ni \forall x,\ 0 < |x - a| < \delta \to |f(x) - L| < \varepsilon$
3. $\forall x$ and y in A, $\exists z$ in $B \ni x + y = z$
4. $\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \ni \forall n,$ if $n \geq N,$ then $\forall x \in S,\ |f_n(x) - f(x)| < \varepsilon$

Example 1.24

Determine the truth value of each statement. Assume that x and y are real numbers. Justify your answer.

1. $\forall x,\ \exists y \ni x + y = 3$
2. $\exists x \ni \forall y,\ x + y \neq 3$
3. $\forall x,\ \exists y \ni xy = 1$
4. $\exists y \ni \forall x,\ xy = 1$

Rules of Inferential Logic

- The main concern of logic is how the truth of some propositions is connected with the truth of another. Thus, we will usually consider a group of related propositions.
- Proofs in mathematics are valid arguments that establish the truth of mathematical statements.
- By an **argument**, we mean a sequence of statements that end with a conclusion.
- By **validity**, it means that the conclusion or final statements of the argument must follow from the truth of the preceding statements, or **premises**, of the argument.
- That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false.

Definition

An **argument** is a set of two or more propositions related to each other in such a way that all but one of them (the **premises**) are supposed to provide support for the remaining one (the **conclusion**).

Logical Inference for Propositional Logic

Definition (Rules of Inference) or **Logical Inference**

A valid argument form that can be used in the demonstration that arguments are valid.

For example: Suppose we know that a statement of the form $p \to q$ is true. This tells us that whenever p is true, q will also be true. By itself, $p \to q$ being true does not tell us that either p or q is true (they could both be false, or p and q be false and q is true).

However, if in addition we happen to know that p is true, then it must be that q is true. This is called a **logical inference**: Given two true statements, we can infer a third statement is true. That is:

$$\begin{array}{c} p \rightarrow q \\ p \\ \hline q \end{array}$$

Other examples are:

$$\begin{array}{c} p \rightarrow q \\ \sim q \\ \hline \sim p \end{array} \qquad \begin{array}{c} p \vee q \\ \sim p \\ \hline q \end{array}$$

- The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the rule of inference called **modus ponens**, or the **law of detachment**.

Table 1.1

	Rules of Inference	Name
1.	p $p \to q$ $\overline{\therefore q}$	Modus Ponens
2.	$\sim q$ $p \to q$ $\overline{\therefore \sim p}$	Modus Tollens
3.	$p \to q$ $q \to r$ $\overline{\therefore p \to r}$	Hypothetical Syllogism
4.	$p \vee q$ $\sim p$ $\overline{\therefore q}$	Disjunctive Syllogism

Table 1.1 (Continued)

	Rules of Inference	Name
5.	$\dfrac{p}{\therefore\ p \vee q}$	Addition
6.	$\dfrac{p \wedge q}{\therefore\ p}$	Simplification
7.	$\dfrac{\begin{array}{c} p \\ q \end{array}}{\therefore\ p \wedge q}$	Conjunction
8.	$\dfrac{\begin{array}{c} p \vee q \\ \sim p \vee r \end{array}}{\therefore\ q \vee r}$	Resolution

Definition (Fallacy)

An invalid argument form often used incorrectly as a rule of inference (or sometimes, more generally, an incorrect argument).

For example, the proposition

$$[(p \to q) \land q] \to p$$

is not a tautology, because it is false when p is false and q is true.

This type of incorrect reasoning is called the **fallacy of affirming the conclusion**.

Example 1.25

Is the following argument valid?
If you do every problem on this slide, then you will learn logic and set theory. You learned logic and set theory.

Solution:

Let p be the proposition "You did every problem in this slide".
Let q be the proposition "You learned Logic and Set theory".

Another example of the proposition

$$[(p \to q) \wedge \sim p] \to \sim q$$

is not a tautology, because it is false when p is false and q is true.

This type of incorrect reasoning is called the **fallacy of denying the hypothesis**.

Example 1.26

Let p and q be as in Example 0.25 If the conditional statement $p \to q$ is true and $\sim p$ is true, is it correct to conclude that $\sim q$ is true?

Rules of Inference for Quantified Statements

- **Universal instantiation**: is the rule used to conclude that $p(c)$ is true, where c is a particular member of the domain, given the premise that $\forall x\ p(x)$.
  For example, "All women are wise" means that "Ama is wise", where Ama is a member of the domain of all women.
- **Universal generalization**: is the rule of inference that states that $\forall x\ p(x)$ is true, given the premise that $p(c)$ is true for all elements c in the domain.
  Universal generalization is used when we show that $\forall x\ p(x)$ is true by taking an arbitrary element c from the domain and showing that $p(c)$ is true.
- **Existential instantiation**: is the rule that allows us to conclude that there is an element c in the domain for which $p(c)$ is true if we know that $\exists x\ p(x)$ is true.
  The value c cannot be selected arbitrarily, but rather it must be a c for which $p(c)$ is true.
  Usually there is no knowledge of what c is, only that it exists.

- **Existential generalization** is the rule of inference that is used to conclude that $\exists x\ p(x)$ is true when a particular element c with $p(c)$ true is known. That is, if we know one element c in the domain for which $p(c)$ is true, then we know that $\exists x\ p(x)$ is true.

Table 1.2: Rules of Inference for Quantified Statements

Rule of Inference	**Name**
$\dfrac{\forall x p(x)}{\therefore\ p(c)}$	Universal instantiation
$\dfrac{p(c) \text{ for an arbitrary } c}{\therefore\ \forall x p(x)}$	Universal generalization
$\dfrac{\exists x p(x)}{\therefore\ p(c) \text{ for some element } c}$	Existential instantiation
$\dfrac{p(c) \text{ for some element } c}{\therefore\ \exists x p(x)}$	Existential generalization

Example 1.27

1. Show that the premises "Everyone in this Logic and Set Theory class has taken a course in Actuarial Science" and "Abena is a student in this class" imply the conclusion "Abena has taken a course in Actuarial Science".

2. Show that the premises "A student in this class has not read the slide" and "Everyone in this class passed the first exam" implying the conclusion "someone who passed the first exam has not read the slide".

Combining Rules and Inference for Propositions and Quantified Statements

- The combination of *universal instantiation* and *modus ponens*, which is used so often together, is sometimes called **universal modus ponens**.
- This rule tells us that if $\forall x\,(p(x) \to q(x))$ is true, and if $p(a)$ is true for a particular element a in the domain of the universal quantifier, then $q(a)$ must also be true. Symbolically:

$$\forall x\,(p(x) \to q(x))$$

$$p(a), \quad \text{where } a \text{ is a particular element in the domain}$$

$$\therefore q(a)$$

Example 1.28

Assume that for all positive integers n, if n is greater than 4, then n^2 is less than 2^n. That is,

$$\forall n \in \mathbb{Z}^+\,(n > 4 \to n^2 < 2^n).$$

Use universal modus ponens to show that

$$100^2 < 2^{100}.$$

Another useful combination of a rule of inference from propositional logic and a rule of inference for quantified statements is *universal modus tollens*.

$$\forall x\,(p(x) \to q(x))$$

$$\sim q(a),\ \text{where } a \text{ is a particular element in the domain}$$

$$\therefore \sim p(a)$$

METHODS OF PROOF

We now explore ways in which ideas from propositional and predicate logic are used to prove mathematical theorems.

Some vocabulary of logic and mathematics would be introduced.
The aim is to be able to read and write mathematics, and this requires more than just vocabulary.
It also requires syntax.That is, we need to understand how statements are combined to form the mysterious mathematical entity known as a poof.

A proof is a valid argument that establishes the truth of a mathematical statement.
A proof can use the hypotheses of the theorem if any, **axioms** assumed to be true, **definitions**, and previously proven theorems.

Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

Some Terminology

- An **axiom:** is a statement that is assumed to be true.
- A **definition :** is used to create new concepts in terms of existing ones.
- A **theorem :** is a proposition that has been proved to be true.
- A **lemma :** is a theorem that is usually not of interest in its own right but is useful in proving theorem.

- **Corollary:** A corollary is a theorem that can be established directly from a theorem that has been proved.
- **Conjecture:** A conjecture is a statement that is proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.
  Alternatively, a **conjecture** is a mathematical assertion proposed to be true ,but has not been proved.
  An argument that establishes the truth of a theorem is called a **proof**.
- A **proof** of as theorem is a written verification that shows that the theorem is definitely and unequivocally true.

Constructive and Non-constructive Proofs

Some mathematical theorems have the form:

$$\exists x \ni \; p(x),$$

while others have the form:

$$\forall x, \; p(x).$$

The theorem of the form:

$$"\exists x \ni p(x)"$$

that is, to prove such a theorem, we try to find a number or element x which has the property $p(x)$.
This theorem guarantees the existence of at least one x for which the predicate $p(x)$ is true.
The proof of such a theorem is **constructive**: that is, the proof is either by finding a particular x that makes $p(x)$ true or by exhibiting an algorithm for finding x.

Example 1.29

1. Show that there exists a positive integer which can be written as the sum of the squares of two numbers.
2. Show that there exists an integer x such that
$$x^2 = 15129.$$
3. Prove that there exists a number x with the property that, for every real number y,$xy = x$.

By a **non-constructive** existence proof we mean a method that involves either showing the existence of x using a proved theorem (or axioms), or showing that the assumption that there is no such x leads to a contradiction.
The disadvantage of a non-constructive method is that it may give virtually no clue about where or how to find x.
Theorems are often of the form
$$"\forall x \in D, \text{ if } p(x) \text{ then } q(x)."$$
We $p(x)$ the hypothesis and $q(x)$ the conclusion.

Let us first consider a proposition of the form

$$\forall x \in D, p(x).$$

Then this can be written in the form
"

$$\text{"}\forall x, \text{ if } x \in D \text{ then } p(x)\text{"}$$

If D is a finite set, then one can check the truth value of $p(x)$ for each $x \in D$. This method is called the **method of exhaustion.**

Example 1.30

Show that for each integer $1 \leq n \leq 10$,

$$n^2 - n + 11$$

is a prime number.

- **Exhaustive proof:** is a proof that establishes a result by checking a list of all cases.
- **Inductive reasoning** is drawing a conclusion on the basis of individual cases.

For Example

If $p(n)$ is the statement
" $n^2 + n + 17$ is a prime number " and we check for the first few positive integers, For example ,

$$p(1) = 19$$
$$p(2) = 23$$
$$p(3) = 29$$
$$p(4) = 37$$

$$p(8) = 89$$

$$p(12) = 173$$

$$p(15) = 257$$

A **counterexample** such as $n = 16 or 17$
i.e. $16^2 + 16 + 17 = 16(16+1) + 17 = 16(17) + 17 = (17)(17)$
or $17^2 + 17 + 17 = 17(17+1) = 17 \cdot 19$
provides that

$$"\forall n\ ,p(n)" \text{ is false.}$$

Example 1.31

Provide counterexamples to the following statements:
1. All birds can fly.

2. Every continuous function is differentiable.

- **Deductive reasoning**is applying a general principle to a specific case.

For example, consider $f(n,m) = n^2 + n + m$, where n and m are understood to be positive integers. We observe that

$$f(1,2) = 1^2 + 1 + 2 = 4 = 2^2$$

$$f(2,3) = 2^2 + 239 = 3^2$$

$$f(5,6) = 5^2 + 5 + 6 = 36 = 6^2$$

$$f(12,13) = 12^2 + 12 + 13 = 169 = 13^2$$

On the basis of these examples (using inductive reasoning) we can form the conjecture

$$\text{“}\forall n \; p(n) \text{”}$$

where $p(n)$ is the statement.”

$$f(n, n+1) = (n+1)^2$$

It turns out that the conjecture this time is true that is

$$f(n, n+1) = n^2 + n + (n+1) = (n+1)^2.$$

It is concluded that

$$\text{“}\forall n \ p(n)\text{”} \text{ is true,}$$

Now that we have proved the general statement

$$\text{“}\forall n \ (n, p(n))\text{”},$$

a particular case without case without computation is

$$f(124, 125) = 125^2,$$

This is an example of **deducative reasoning.**

SUMMARY TABLE

Constructive Proof	**Non-constructive Proof**
Gives an explicit example	Does not give an explicit example
Shows how to build the object	Shows the object must exists
More "hands on"	More "logical"
Often preferred in algorithms	Common in pure mathematics

Methods of Proving Theorems

Proving theorems can be difficult. We need all the ammunition that is available to help us prove different results. This last part of logic introduces a battery of different proof methods.

To prove a theorem of the form

$$\forall x, p(x) \to q(x),$$

Our goal is to show that

$$p(a) \to q(a)$$

is true, where a is an arbitrary element of the domain, and then apply universal generalization.

Direct Proof

Direct Proof of a conditional statement

$$p \rightarrow q$$

is considered when the first step is the assumption that p is true; subsequent steps are considered using rules of inference with the final step showing that q must also be true.
A direct proof shows that a conditional statement

$$p \rightarrow q$$

is true by showing that if p is true, then q must also be true, so that the combination p true and q false never occurs.

In a direct proof, we assume that p is true and use axioms, definitions, and previously proven theorems together with rules of inference to show that q must also be true.

Example 1.32

Prove that:

1. If n is an odd integer, then n^2 is odd.
2. If n and m are both perfect squares, then nm is also a perfect square.

Example 1.33

Prove the following theorem:

1. For every $\varepsilon > 0$ there exists a $\delta > 0$ such that 1 - $\delta < x <$1 + δ implies that $5 - \varepsilon < 2n + 3 < 5 + \varepsilon$.
2. Let x and y be positive numbers.
   If $x \leq y$, then $\sqrt{x} \leq \sqrt{y}$.
3. If x and y are positive real numbers, then $2\sqrt{xy} \leq x + y$.

Indirect Proofs

Direct proof leads from the hypothesis of a theorem to the conclusion. They begin with premises, continue with a sequence of deductions, and end with the conclusion.

However, some theorems with direct proofs often reach dead ends.

Other methods of proving theorems of the form:

$\forall, p(x) \rightarrow q(x)$

are not direct proofs , that is, that don't start with the hypothesis and end with the conclusion are called **indirect proofs**.

Two main types of indirect proofs:

- Proof by contraposition
- Proof by contradiction

Proofs by contraposition

Proofs by contraposition make use of the fact that the conditional statement

$$p \to q$$

is equivalent to its contrapositive,

$$\sim q \to \sim p$$

This means that the conditional statement $p \to q$ can be proved by showing that its contrapositive

$$\sim q \to \sim q$$

is true.

Example 1.34

1. Prove that if n is an integer and $3n + 2$ is odd, then m^n is odd.
2. Prove that if $n = ab$, where $a and b$ are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$

Example 1.35

Prove the following theorems:

1. Let f be an integrable function. If

$$\int_0^1 f(x)\,dx \neq 0,$$

   then there exists a point x in the interval $[0, 1]$ such that $f(x) \neq 0$.
2. Let x be a real number. If $x > 0$, then $x > 0$.
3. Suppose $x \in \mathbb{Z}$, if $7x + 9$ is even, then x is odd.
4. Suppose $x \in \mathbb{R}$, if $x^2 - 6x + 5$ is even, then x is odd.
5. Suppose $x, y \in \mathbb{R}$), if $y^3 + yx^2 \leq x^3 + xy^2$, then $y \leq x$.

Vacuous and Trivial Proofs

We can quickly prove that a conditional statement $p \rightarrow q$ is true when we know that p is false, because $p \rightarrow q$ must be true when p is false.

Vacuous Proof is a proof that

$$p \rightarrow q$$

is true based on the fact that p is false.

Example 1.36

Show that the proposition $p(0)$ is true, where $p(n)$ is

$$\text{"if } n > 1, \text{ then } n^2 > n\text{"}$$

and the domain consists of all integers.
We can also quickly prove a conditional statement $p \rightarrow q$ if we know that the conclusion q is true,
By showing that p is true, it follows that $p \rightarrow q$ must also be true.

Trivial proof is a proof that

$$p \to q$$

is true based on the fact that q is true.

Example 1.37

Let $p(n)$ be "if a and b are positive integers, with $a \geq b$, then $a^2 \geq b$, where the domain consists of all integers.Show that $p(0)$ is true.

Proofs by Contradiction

Definition. A proof that p is true based on the truth of the conditional statement

$$\sim p \to \sim q, \text{where}$$

q is a contradiction.
The statement $(r \sim r)$ is a contradiction. Whenever r is a proposition, we can prove that p is true if we can show that

$$\sim p \to (r \sim r)$$

is true for some proposition r.

Example 1.38

1. Prove that $\sqrt{2}$ is an irrational number.
2. Prove that "if $3n + 2$ is odd, then n is odd".
3. Let x be a real number. If $x > 0$, then $\frac{1}{x} > 0$.
4. Prove that if x is a real number, then $x \leq |x|$.

Exercises

1. Mark each statement True or False. Justify each answer.
   a. The symbol "$\exists$" means "there exist several."
   b. If a variable is used in the antecedent of an implication without being quantified, then the universal quantifier is assumed to apply.
   c. The order in which quantifiers are used affects the truth value.

2. Write the negation of each statement.
   a. Everyone likes Robert.
   b. Some students work part-time.
   c. No square matrices are triangular.
   d. $\exists x$ in $B \ni f(x) > k$.
   e. If $x > 5$, then $f(x) < 3$ or $f(x) > 7$.
   f. If x is in A, then $\exists y$ in $B \ni f(x) < f(y)$.

3. Determine the truth value of each statement, assuming x is a real number. Justify your answer.
   a. $\exists x$ in the interval $[2, 4] \ni x < 7$.
   b. $\forall x$ in the interval $[2, 4], x < 7$.
   c. $\exists x \ni x^2 = 5$.
   d. $\forall x, x^2 = 5$.
   e. $\exists x \ni x^2 \neq -3$.
   f. $\forall x, x^2 \neq -3$.
   g. $\exists x \ni x \div x = 1$.
   h. $\forall x, x \div x = 1$.
4. Determine the truth value of each statement, assuming x is a real number. Justify your answer.
   a. $\exists x$ in the interval $[3, 5] \ni x \geq 4$.
   b. $\forall x$ in the interval $[3, 5], x \geq 4$.
   c. $\exists x \ni x^2 \neq 3$.
   d. $\forall x, x^2 \neq 3$.
   e. $\exists x \ni x^2 = -5$.
   f. $\forall x, x^2 = -5$.
   g. $\exists x \ni x - x = 0$.
   h. $\forall x, x - x = 0$.

5. Below are two strategies for determining the truth value of a statement involving a positive number x and another statement $P(x)$:
   1. Find some $x > 0$ such that $P(x)$ is true.
   2. Let c be the name for any number greater than 0 and show $P(x)$ is true.

   For each statement below, indicate which strategy is more appropriate.

   a. $\forall x > 0, P(x)$.
   b. $\exists x > 0 \ni P(x)$.
   c. $\exists x > 0 \ni \sim P(x)$.
   d. $\forall x > 0, \sim P(x)$.